

Haychan Shin

Email: shinhaychan@gmail.com | Portfolio: charlieshin.me | Github: <https://github.com/charshinbyte>

EDUCATION

University of Cambridge, Gonville and Caius College

Graduation Date: June 2026

MEng, BA Electrical Engineering and Information sciences

- **Relevant Modules:** linear algebra, probability theory, object-oriented programming, statistics, stochastic process, MLE, information theory, data transmission, signals and systems, computer vision, probabilistic machine learning

SKILLS

Languages: Python, JavaScript, Octave, HTML, CSS, SQL, C, C++

Frameworks & Technologies: PyTorch, Tensorflow, HuggingFace, NumPy, pandas, Matplotlib, Seaborn, Flask, PyQt5, FastAPI, Git, Docker, Postman, MongoDB Atlas, Motor, SQLite, AWS Lambda

WORK EXPERIENCES

Research Assistant | Yonsei University Medical AI Lab

July 2025 - September 2025

- Advised under Professor Hwang Dosik researching domain adaptation for MRI tumour classification

3D Reconstruction Researcher | Republic of Korea Army

December 2023 - May 2025

Python • Pytorch • FastAPI • Neural Radiance Fields • 3D Gaussian Splat

- Designed a platform for improved depth estimations under noisy illuminated images using Neural Radiance Fields.
- Developed novel deep-learning pretraining scheme for enhanced robustness of implicit neural representation under minimal training data, achieving 10% improved SSIM scores.

Electrical Engineering Intern | TTP plc

July 2023 - September 2023

Python • PCB Design • Altium Designer • Signal Processing

- Spearheaded the development of a dynamic ray-traced model in Python for a novel optical alignment system culminating in **60% improvement in detection area** and **15% improved precision**.
- Designed, populated and tested PCB designs for sensor arrays, evaluating accuracy and optical reflection properties.

PROJECTS (charlieshin.me)

DJ Y - LLM based Assistant for Music Recommendation

Web Scraping • RESTful APIs • LLMs • CNNs • TTS (ElevenLabs)

- Built an AI-driven assistant that recommends music by analyzing lyrics sentiment and audio features using custom-trained CNN models.
- Integrated Spotify Web API with real-time web scrapers to provide personalized listening experience alongside contextual information such as weather and trending news.
- Improved user engagement through natural voice interaction, prompt-tuned Gemini LLMs, and ElevenLabs TTS for seamless spoken dialogue.

Shazam Clone - Music Identifier (<https://shorturl.at/VbWLA>)

Python • AWS • RESTful APIs • MongoDB

- Engineered a high-accuracy music recognition system that identifies songs from short audio clips using a custom **frequency–time hashing algorithm with O(1) lookup efficiency**, achieving **78% accuracy** in real world noisy environments.
- Developed and deployed a scalable, containerized API to AWS EC2 for song matching and library updates, leveraging **MongoDB Atlas** for fingerprint storage and integrating **SQL** for metadata mapping, enabling optimized retrieval across hybrid storage systems.

REFERENCES

Dr Steven Jones - Former Employer: sjones@cadfem.co.uk

Dr Malcolm Smith - Director of Studies: mcs1000@cam.ac.uk